

KAHULUI AP, HI, USA

WMO: 911900

Lat: 20.900N

Lon: 156.429W

Elev: 51

StdP: 14.67

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 22516

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	59.2	60.9	52.2	58.1	67.3	54.2	62.6	67.6	29.4	77.6	26.8	76.6	4.5	150	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.9	90.3	75.0	88.9	74.5	87.8	74.0	77.5	86.0	76.5	85.0	75.8	84.3	17.4	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.2	132.5	80.9	74.1	127.4	80.5	73.1	123.2	80.1	40.9	86.0	39.9	85.2	39.1	84.4	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.8	25.2	23.9	DB	54.3	92.6	2.7	2.1	52.4	94.2	50.9	95.4	49.4	96.6	47.4	98.1	
			WB	52.1	78.6	2.2	2.4	50.6	80.3	49.3	81.7	48.1	83.1	46.5	84.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	76.1	72.1	72.2	73.0	74.6	76.0	78.1	79.4	79.9	79.5	78.3	75.9	73.6
	DBStd	3.61	2.58	2.52	2.44	2.43	2.16	1.80	1.95	1.84	2.21	2.14	2.36	2.36
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9516	685	620	712	738	807	844	910	927	885	879	778	731
	CDD65	4041	220	200	247	288	342	394	445	462	435	414	328	266
	CDH74	29656	1136	1000	1286	1726	2470	3143	3850	4145	3967	3424	2140	1369
	CDH80	8224	119	110	177	345	648	972	1279	1425	1395	1095	484	175
Wind	WSAvg	12.3	10.4	10.8	11.5	12.8	12.2	14.5	14.8	14.4	12.7	11.8	11.3	10.8
Precipitation	PrecAvg	17.0	3.4	2.5	2.6	1.3	0.7	0.2	0.5	0.5	0.4	1.0	2.3	2.9
	PrecMax	34.7	14.4	8.3	10.9	4.4	3.0	0.8	1.7	2.4	1.4	5.7	9.3	10.2
	PrecMin	5.3	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0
	PrecStd	7.5	3.3	2.2	2.3	1.2	0.8	0.2	0.4	0.5	0.4	1.1	2.1	2.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.9	84.8	85.4	87.1	89.2	90.1	91.4	91.3	91.7	90.9	88.4	85.7
		MCWB	71.0	71.4	71.3	72.3	73.4	73.8	75.6	76.3	75.6	75.1	74.4	72.6
	2%	DB	82.7	82.8	83.4	85.1	86.6	87.8	89.2	89.4	89.8	88.8	86.2	83.5
		MCWB	70.3	70.8	70.8	71.6	72.4	73.0	74.8	74.9	75.0	74.6	73.4	71.3
	5%	DB	81.3	81.4	82.1	83.5	85.2	86.4	87.5	88.0	88.2	87.3	84.5	82.0
		MCWB	70.0	70.0	70.3	71.1	71.6	72.5	74.0	74.3	74.4	74.0	72.8	70.7
	10%	DB	79.9	79.8	80.6	82.1	83.7	85.1	86.1	86.7	86.8	85.7	82.9	80.6
		MCWB	69.4	69.3	69.7	70.5	71.3	72.1	73.3	73.8	73.8	73.3	72.2	70.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.3	74.6	75.0	76.2	76.0	76.1	78.6	78.9	78.9	77.7	76.8	75.7
		MCDB	79.9	81.0	81.1	81.8	84.1	86.1	86.3	87.6	87.6	86.4	84.2	81.7
	2%	WB	73.0	72.9	73.2	73.5	74.6	75.0	76.8	77.2	77.2	76.4	75.4	73.7
		MCDB	78.9	79.3	79.6	81.1	83.2	84.7	85.2	86.0	85.9	85.0	82.8	79.9
	5%	WB	71.8	71.7	71.9	72.4	73.4	74.1	75.6	76.1	76.1	75.5	74.3	72.5
		MCDB	78.2	78.3	78.7	80.5	81.9	83.7	84.2	84.8	85.0	83.9	81.6	78.8
	10%	WB	70.7	70.7	70.9	71.3	72.5	73.2	74.6	75.2	75.1	74.6	73.2	71.3
		MCDB	77.3	77.4	77.8	79.8	81.0	82.7	83.3	83.7	84.1	83.0	80.4	77.8
Mean Daily Temperature Range	MDBR		16.0	15.7	15.2	15.2	16.2	15.3	14.8	14.9	15.9	15.9	14.9	14.9
	5% DB	MCDBR	18.3	18.5	18.0	17.9	18.6	17.0	16.5	16.3	17.4	17.8	17.4	17.4
		MCWBR	9.6	9.5	9.0	8.3	8.4	7.3	7.1	7.1	7.6	8.1	8.5	9.1
	5% WB	MCDBR	15.8	15.7	16.1	16.3	16.8	15.8	15.3	15.5	16.4	16.8	15.9	14.8
		MCWBR	8.8	8.6	8.8	8.3	8.2	7.4	7.1	7.4	7.7	8.5	8.5	8.6
Clear-Sky Solar Irradiance	taub		0.390	0.398	0.410	0.423	0.413	0.398	0.399	0.399	0.389	0.400	0.401	0.382
	taud		2.470	2.445	2.419	2.364	2.429	2.471	2.478	2.478	2.505	2.487	2.473	2.519
	Ebn at Noon		280	284	284	280	280	282	282	284	286	280	274	279
	Edh at Noon		33	35	37	40	37	36	35	35	34	33	32	30
All-Sky Solar Radiation	RadAvg		1422	1602	1795	2049	2182	2245	2220	2156	2019	1710	1444	1301
	RadStd		60	95	167	121	112	58	54	87	57	44	35	111

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	+351	+349
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	+162

Nomenclature: See separate page